

# Luis Eduardo González González

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Portfolio

## PROFESSIONAL SUMMARY

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IT Professional with extensive experience in full-stack development, cloud software architecture, and technology project management. Specialized in Python, JavaScript, and Google Cloud solutions. Recently focused on high-level independent projects involving local generative AI orchestration, advanced observability, and distributed systems. Passionate about building efficient, high-quality solutions, backed by a strong analytical foundation from a background in Physical Sciences.

## TECHNICAL SKILLS

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- Programming Languages: Python, JavaScript, Java, Go, Fortran, C, PHP
- Web Development (Front-End): HTML5, CSS3 (SASS), JavaScript, Vue.js, jQuery, Bootstrap, Materialize
- Web Development (Back-End): Python (Flask), PHP (Laravel), Node.js
- AI & Automation (New Additions): Prompt Engineering (Expert Systems), AI Agents (OpenCode, Gemini-CLI), Local LLMs (Ollama, GPU optimization), n8n (AI Automation), Power BI
- DevOps & Cloud: Google Cloud Platform (GCP), AWS (Basics), Docker & Docker Compose (GPU Support), Kubernetes (K8s), Terraform, Ansible, Xen Orchestra
- Infrastructure & Networking: Cloudflare Tunnel, Zero Trust, Kafka, RabbitMQ
- Observability: Grafana, Istio, Prometheus, Traceability, Video Coding
- Databases: SQL (PostgreSQL, MySQL, MariaDB), NoSQL (MongoDB, Redis)
- Version Control: Git
- Operating Systems: Linux (Debian), macOS, Windows
- Tools & Others: Arduino, Bash/Shell Scripting, Agile Methodologies (Scrum), Frigate
- Data Analysis/Simulation: Matlab, Mathematica, SPSS, Origin

## PROFESSIONAL EXPERIENCE

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### 3.1 Solutions Architect & Maker | Independent Consulting

*Jan 2026–Present*

- **Infrastructure:** Orchestration of on-premise servers (Xen Orchestra, K8s, RabbitMQ).
- **Applied AI:** Implementation of local LLMs (Ollama) for educational and creative use cases.
- **Maker:** Design and fabrication of custom hardware (3D Printing, IoT).

### 3.2 Research and Development (R&D) Lead | GET IN, S.A.P.I. de C.V.

*Jan 2024–Jan 2026*

- In charge of executing proofs of concept to evaluate the technical and functional viability of new solutions and innovations.
- Completely designed and developed a statistical model for automating data review, resulting in a 90% reduction in manual processing times.
- Continuously formulated hypotheses and developed proofs of concept to validate new technological approaches, actively collaborating with development teams for implementation.

### 3.3 Chief Information Officer (CIO) | GET IN, S.A.P.I. de C.V.

*Jun 2018–Jan 2024*

- Led technology strategy and IT teams, covering product feature development, inventory management, support, and process design.
- Directed the infrastructure migration to **Google Cloud Platform (GCP)** implementing PaaS solutions, significantly reducing maintenance costs and increasing service reliability.
- Driven the re-engineering of system architecture towards a microservices model, facilitating evolutionary maintenance and scalability.
- Redesigned data processing systems into a horizontal structure, cutting processing times to one-third and reducing associated operating costs by one-third.
- Managed 5 multidisciplinary teams: Front-End, Back-End, Embedded Software, Data Engineering, and Warehouse.
- Responsible for requirements definition, staff technical training, and critical incident resolution.
- Key Technologies: Go, Python, PHP (Laravel), MongoDB, Java, Vue.js, GCP.

### 3.4 Senior Developer | elcultivo.mx

*Feb 2015–Jun 2018*

- Transformed the company's focus from portfolio web design to custom CRM application development for e-commerce.
- Established and led new development processes, successfully delivering 20 distinct web applications for 20 different clients in one year.
- Responsible for software architecture design (ER Models, MVC with **Laravel** and **Vue.js**).
- Led the development team, managing requirements gathering and technical training.
- Collaborated closely with graphic designers on UI/UX optimization.
- Key Technologies: PHP (Laravel), Vue.js, JavaScript, MySQL, MongoDB, HTML5, CSS3, Redis.

### 3.5 PAIMEF Monitoring Liaison | National Institute for Social Development (Indesol)

*Jan 2013–Jan 2015*

- Responsible for generating technical mechanisms for evaluating and monitoring PAIMEF resources.
- Identified inefficiencies in KPI tracking (done via nested spreadsheets) and proposed a centralized web system for data capture and analysis.
- Proactively designed and developed two interconnected web systems (PHP/MySQL) to automate data collection, database storage, and indicator calculation.
- Implemented and maintained successive versions of these systems, significantly improving monitoring efficiency and reliability.

## TEACHING EXPERIENCE

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### 4.1 Guest Lecturer, Master in Web Development | Universidad Andina Simón Bolívar

*Jan 2018–Feb 2018*

- Designed and taught a specialized module on Object-Relational Mappers (ORM), with a practical focus on Eloquent (Laravel's ORM).
- Responsible for creating teaching materials and conducting sessions, receiving positive evaluations from students.

## EDUCATION

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### 5.1 Diploma in Results-Based Budgeting (PbR) | UNAM / SHCP (Ministry of Finance)

*Jan 2014–Dec 2014*

- Theoretical basis for the design of monitoring systems at Indesol.

### 5.2 M.S. in Physics | UNAM (National Autonomous University of Mexico)

*Jan 2008–Jan 2010*

- Specialization in Mesoscopic Nonequilibrium Thermodynamics.
- Study of complex systems and statistical fluctuations.
- Mathematical modeling, computational simulation, scientific rigor.

### 5.3 B.S. in Physics | UNAM (National Autonomous University of Mexico)

*Jan 2003–Jan 2007*

- Fundamentals: Quantum Mechanics, Electromagnetism, Statistical Mechanics.

## INDEPENDENT PROJECTS & FREELANCE

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### Infrastructure, Cloud & Observability (2026 - Present)

- Local Server Orchestration: Management of on-premise infrastructure using Xen Orchestra, Terraform, and Ansible for automated provisioning.
- Scalable Distributed Systems: Development of a Kubernetes (K8s) message processing system integrated with RabbitMQ. Implementation of pod autoscaling based on workload and pipelines where each worker runs independent jobs.
- Security & Networking: Implementation of Cloudflare Tunnel and Zero Trust policies for secure resource access.
- Advanced Observability: Deployment of monitoring stacks with Grafana, Istio, and Prometheus; implementation of distributed tracing for microservices.

### Generative AI & Automation

- Language Learning Platform (AI Roleplay): Development of a local dockerized application utilizing LLMs (via Ollama with GPU optimization) to simulate a language instructor. The system manages complex role-play interactions via voice and text.
- AI Agents & Workflows: Creation and programming of autonomous agents using OpenCode and Gemini-CLI. Automation of complex business processes using intelligent workflows in n8n.
- Advanced Prompt Engineering: Design of "Expert Systems" using advanced prompting techniques for specialized tasks.

### Simulations & Home Automation (Historical)

- Physical Simulations: Developed interactive simulations for planetary motion (Basic) and gravitational waves (Fortran).
- Neural Networks: Experimented with development and training of neural networks for image classification using Python/TensorFlow.
- Advanced Home Automation: Implemented control systems (lighting, temperature, security) with Home Assistant and Frigate.
- 3D Design: Architectural and technical part modeling for 3D printing (SketchUp, FDM).

## LANGUAGES

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Spanish: Native | English: B2 Level (Professional Working Proficiency)

## PUBLICATIONS / OTHER

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- Co-author: "Experiments with Soap Bubbles", FENOMECE Notes (science outreach publication) (2010) — Publication
- Electrical Technician skills — Certification
- PC and electronics repair
- Advanced proficiency in office suites (Microsoft Office Suite, Google Workspace)